

Sanibel v2.0.1

Advanced Molecular Detection Southeast Region

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OVERVIEW

Purpose:

- This Nextflow pipeline processes whole sequencing reads and paired-end WGS data to perform QC, assembly, taxonomic classification, species identification, sequencing typing, AMR detection and run automated species-specific typing modules along with plasmid and enhanced AMR analysis tools.

Usage:

- To help public health labs and researchers by analyzing sequencing data and generating results for surveillance, outbreak investigations, and routine molecular typing.

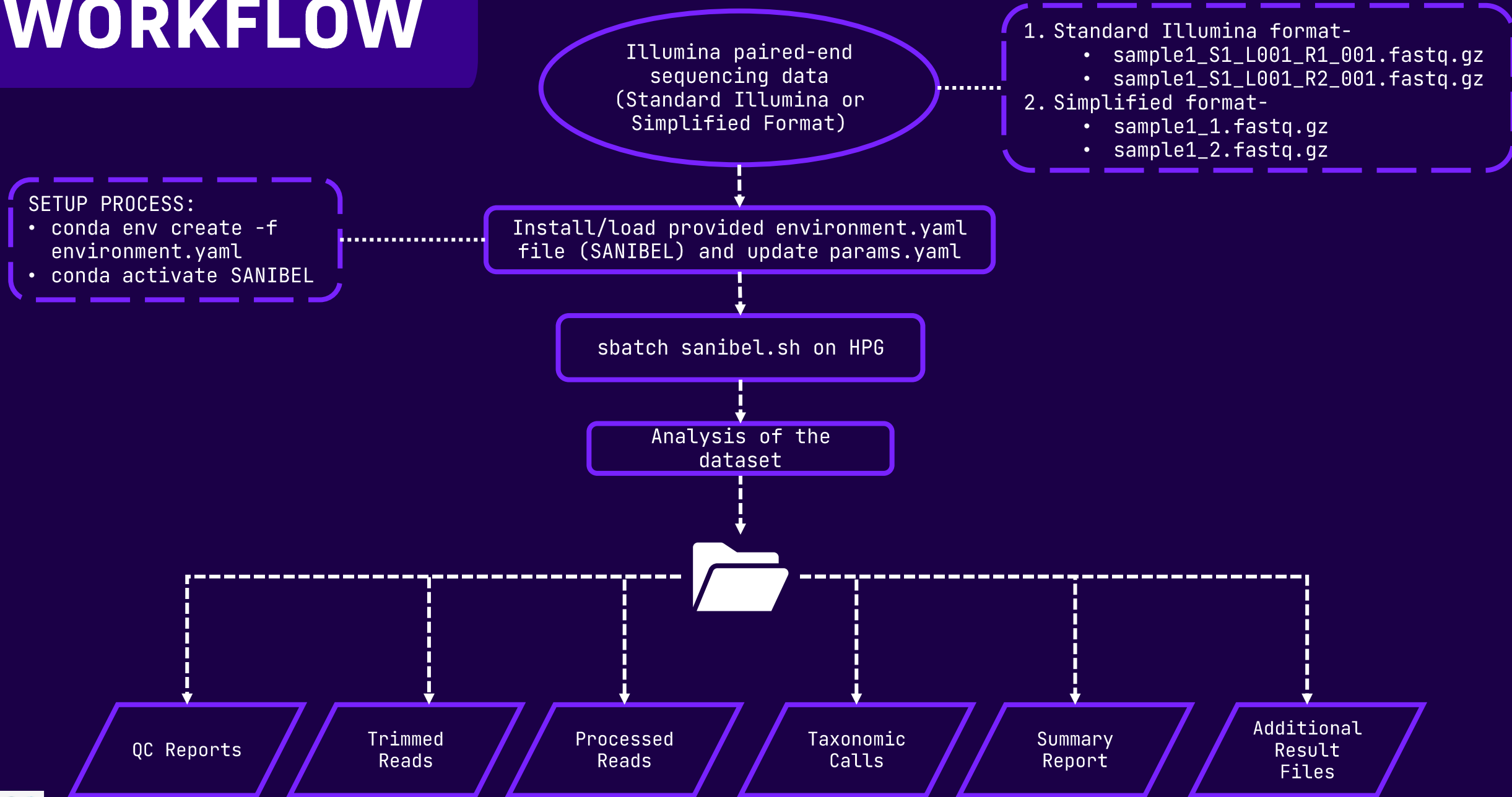
Dependencies:

- Nextflow
- Conda
- Apptainer/Singularity
- SLURM workload manager (required for HiPerGator; otherwise not required)

WHAT IS NEW IN SANIBEL 2.0.1?

Feature/Version Change	Sanibel 1.3.0	Sanibel 2.0.1
Core Pipeline	Nextflow NGS Pipeline + QC, Assembly, Typing, AMR	Full rewrite in DSL2, improved modules & resource management
Species Modules	<i>Neisseria</i> , <i>H. influenzae</i> , <i>Legionella</i> , <i>Shigella</i> , GAS, <i>Klebsiella</i> , <i>Salmonella</i> , <i>E. coli</i> , plasmids	Added SeroBA (<i>Streptococcus pneumoniae</i>), Pasty (<i>Pseudomonas aeruginosa</i>), Kaptive AB/VP (<i>Acetivobacter baumannii</i> , <i>Vibrio parahaemolyticus</i>)
BMGAP2 Integration	Not Included	Added modules for <i>N. meningitidis</i> / <i>H. influenzae</i>
MASH Underscore Bug	Bug existed; underscore genus names broke Prokka	Fixed fully in v2.0 via <code>parse_assembly.py</code>
Summary Report System	Basic reporting	Completely rewritten summary system
Performance & Resource Usage	Baseline runtime	Enhanced SLURM, faster assemblies build
Tool Container Versions	Earlier tool versions	Large-scale version upgrades
Conda Environment	Basic environmental setup	Reproducible Conda env
Nextflow Version	Not limited	Nextflow 25.10.4
Documentation	Basic doc	Enhanced external setup docs

WORKFLOW



APPLICATION

OBJECTIVE

Use an Illumina paired-end dataset (*E. coli*, *S. enterica*, etc.) and analyze the dataset using Sanibel pipeline

APPLICATION CONT.

```
cd /blue/bphl-<state>/<user>/repos/bphl-molecular/  
git clone https://github.com/BPHL-Molecular/Sanibel  
mkdir analysis/  
cd analysis/  
cp /blue/bphl-<state>/<user>/repos/bphl-molecular/Sanibel/*
```

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/Sanibel0727/

Name	Size	Changed	Rights	Owner
..		6/24/2026 12:05:38 PM	rw-r--r--	n.yengalareddy
assets		6/24/2026 10:51:53 AM	rw-r--r--	n.yengalareddy
bin		6/24/2026 10:51:53 AM	rw-r--r--	n.yengalareddy
db		6/24/2026 10:51:53 AM	rw-r--r--	n.yengalareddy
fastqs		6/24/2026 11:54:55 AM	rw-r--r--	n.yengalareddy
licenses		6/24/2026 10:51:54 AM	rw-r--r--	n.yengalareddy
modules		6/24/2026 10:51:54 AM	rw-r--r--	n.yengalareddy
output-202606241...		6/24/2026 1:25:01 PM	rw-r--r--	n.yengalareddy
work		6/24/2026 1:25:00 PM	rw-r--r--	n.yengalareddy
CHANGELOG.md	12 KB	6/24/2026 10:51:53 AM	rw-r--r--	n.yengalareddy
environment.yaml	3 KB	6/24/2026 10:51:53 AM	rw-r--r--	n.yengalareddy
LICENSE	12 KB	6/24/2026 10:51:53 AM	rw-r--r--	n.yengalareddy
nextflow.config	6 KB	6/24/2026 10:51:54 AM	rw-r--r--	n.yengalareddy
params.yaml	1 KB	6/24/2026 12:06:58 PM	rw-r--r--	n.yengalareddy
README.md	10 KB	6/24/2026 10:51:53 AM	rw-r--r--	n.yengalareddy
sanibel.35647270.err	1 KB	6/24/2026 12:22:40 PM	rw-r--r--	n.yengalareddy
sanibel.35647270.out	441 KB	6/24/2026 1:25:01 PM	rw-r--r--	n.yengalareddy
sanibel.nf	9 KB	6/24/2026 10:51:54 AM	rw-r--r--	n.yengalareddy
sanibel.sh	1 KB	6/24/2026 12:21:06 PM	rw-r--r--	n.yengalareddy

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/Sanibel0727/fastqs/*.*

Name	Size	Changed	Rights	Owner
..		6/24/2026 1:25:01 PM	rw-r--r--	n.yengalareddy
sample_data		6/24/2026 11:52:32 AM	rw-r--r--	n.yengalareddy
J4-FL-M08815-240...	281...	6/24/2026 11:52:25 AM	rw-r--r--	n.yengalareddy
J4-FL-M08815-240...	309...	6/24/2026 11:52:25 AM	rw-r--r--	n.yengalareddy
J7-FL-M08815-240...	183...	6/24/2026 11:52:29 AM	rw-r--r--	n.yengalareddy
J7-FL-M08815-240...	201...	6/24/2026 11:52:29 AM	rw-r--r--	n.yengalareddy
J8-FL-M08815-240...	239...	6/24/2026 11:52:30 AM	rw-r--r--	n.yengalareddy
J8-FL-M08815-240...	248...	6/24/2026 11:52:30 AM	rw-r--r--	n.yengalareddy
J9-FL-M08815-240...	219...	6/24/2026 11:52:31 AM	rw-r--r--	n.yengalareddy
J9-FL-M08815-240...	254...	6/24/2026 11:52:31 AM	rw-r--r--	n.yengalareddy
J17-FL-M08815-24...	206...	6/24/2026 11:52:18 AM	rw-r--r--	n.yengalareddy
J17-FL-M08815-24...	225...	6/24/2026 11:52:18 AM	rw-r--r--	n.yengalareddy
J36-FL-M08815-24...	222...	6/24/2026 11:52:19 AM	rw-r--r--	n.yengalareddy
J36-FL-M08815-24...	241...	6/24/2026 11:52:20 AM	rw-r--r--	n.yengalareddy
J40-FL-M08815-24...	143...	6/24/2026 11:52:21 AM	rw-r--r--	n.yengalareddy
J40-FL-M08815-24...	154...	6/24/2026 11:52:22 AM	rw-r--r--	n.yengalareddy
J52-FL-M08815-24...	250...	6/24/2026 11:52:26 AM	rw-r--r--	n.yengalareddy
J52-FL-M08815-24...	293...	6/24/2026 11:52:26 AM	rw-r--r--	n.yengalareddy
J100-FL-M08815-2...	180...	6/24/2026 11:52:14 AM	rw-r--r--	n.yengalareddy
J100-FL-M08815-2...	203...	6/24/2026 11:52:15 AM	rw-r--r--	n.yengalareddy
J102-FL-M08815-2...	182...	6/24/2026 11:52:16 AM	rw-r--r--	n.yengalareddy
J102-FL-M08815-2...	208...	6/24/2026 11:52:17 AM	rw-r--r--	n.yengalareddy
J120-FL-M08815-2...	197...	6/24/2026 11:52:17 AM	rw-r--r--	n.yengalareddy
J120-FL-M08815-2...	217...	6/24/2026 11:52:18 AM	rw-r--r--	n.yengalareddy
J217-FL-M08815-2...	131...	6/24/2026 11:52:19 AM	rw-r--r--	n.yengalareddy
J217-FL-M08815-2...	148...	6/24/2026 11:52:19 AM	rw-r--r--	n.yengalareddy
J414-FL-M08815-2...	250...	6/24/2026 11:52:23 AM	rw-r--r--	n.yengalareddy
J414-FL-M08815-2...	277...	6/24/2026 11:52:24 AM	rw-r--r--	n.yengalareddy
J756-FL-M08815-2...	163...	6/24/2026 11:52:27 AM	rw-r--r--	n.yengalareddy
J756-FL-M08815-2...	196...	6/24/2026 11:52:27 AM	rw-r--r--	n.yengalareddy
J762-FL-M08815-2...	119...	6/24/2026 11:52:27 AM	rw-r--r--	n.yengalareddy
J762-FL-M08815-2...	121...	6/24/2026 11:52:28 AM	rw-r--r--	n.yengalareddy
J780-FL-M08815-2...	148...	6/24/2026 11:52:28 AM	rw-r--r--	n.yengalareddy
J780-FL-M08815-2...	147...	6/24/2026 11:52:28 AM	rw-r--r--	n.yengalareddy

APPLICATION CONT.

```
GNU nano 5.6.1 params.yaml
# Input / Output absolute paths, no trailing slash "/"
input: "/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/Sanibel0727/fastqs"
output: "/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/Sanibel0727/output"

# non-HiPerGator users: uncomment and set your BMGAP2 analysis_scripts directory
# bmgap2_db: "/full/path/to/bmgap2/analysis_scripts"

# non-HiPerGator users: uncomment and set your Kraken2 database directory
# kraken_db: "/full/path/to/kraken2/database"
```

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/Sanibel0727/				
Name	Size	Changed	Rights	Owner
..		6/24/2026 12:05:38 PM	rw-r--r--	n.yengalareddy
assets		6/24/2026 10:51:53 AM	rw-r--r--	n.yengalareddy
bin		6/24/2026 10:51:53 AM	rw-r--r--	n.yengalareddy
db		6/24/2026 10:51:53 AM	rw-r--r--	n.yengalareddy
fastqs		6/24/2026 11:54:55 AM	rw-r--r--	n.yengalareddy
licenses		6/24/2026 10:51:54 AM	rw-r--r--	n.yengalareddy
modules		6/24/2026 10:51:54 AM	rw-r--r--	n.yengalareddy
output-202606241...		6/24/2026 1:25:01 PM	rw-r--r--	n.yengalareddy
work		6/24/2026 1:25:00 PM	rw-r--r--	n.yengalareddy
CHANGELOG.md	12 KB	6/24/2026 10:51:53 AM	rw-r--r--	n.yengalareddy
environment.yaml	3 KB	6/24/2026 10:51:53 AM	rw-r--r--	n.yengalareddy
LICENSE	12 KB	6/24/2026 10:51:53 AM	rw-r--r--	n.yengalareddy
nextflow.config	6 KB	6/24/2026 10:51:54 AM	rw-r--r--	n.yengalareddy
params.yaml	1 KB	6/24/2026 12:06:58 PM	rw-r--r--	n.yengalareddy
README.md	10 KB	6/24/2026 10:51:53 AM	rw-r--r--	n.yengalareddy
sanibel.35647270.err	1 KB	6/24/2026 12:22:40 PM	rw-r--r--	n.yengalareddy
sanibel.35647270.out	441 KB	6/24/2026 1:25:01 PM	rw-r--r--	n.yengalareddy
sanibel.nf	9 KB	6/24/2026 10:51:54 AM	rw-r--r--	n.yengalareddy
sanibel.sh	1 KB	6/24/2026 12:21:06 PM	rw-r--r--	n.yengalareddy

APPLICATION CONT.

```
GNU nano 5.6.1                                sanibel.sh
#!/usr/bin/bash
#SBATCH --account=bphl-umbrella
#SBATCH --qos=bphl-umbrella
#SBATCH --job-name=sanibel
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=40
#SBATCH --mem=200gb
#SBATCH --time=48:00:00
#SBATCH --output=sanibel.%j.out
#SBATCH --error=sanibel.%j.err
#SBATCH --mail-user=Nikhil.Yengala@flhealth.gov
#SBATCH --mail-type=FAIL,END

module load conda apptainer nextflow/25.10.4
conda activate SANIBEL

# Path to container image cache directory
export NXF_APPTAINER_CACHEDIR=/blue/bphl-florida/n.yengalareddy/cache/Apptainer_Singularity/

# Run pipeline
nextflow run sanibel.nf -profile apptainer -params-file params.yaml

# Rename output directory with timestamp
dt=$(date +%Y%m%d%H%M%S")
output_dir=$(grep '^output:' params.yaml | sed 's/output:[[:space:]]*/' | tr -d '"')
mv "$output_dir" "${output_dir}-${dt}"

# Cleanup (disabled for troubleshooting runs)
#rm -rf ./work
```

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/Sanibel0727/

Name	Size	Changed	Rights	Owner
..		6/24/2026 12:05:38 PM	rw-r--r--	n.yengalareddy
assets		6/24/2026 10:51:53 AM	rw-r--r--	n.yengalareddy
bin		6/24/2026 10:51:53 AM	rw-r--r--	n.yengalareddy
db		6/24/2026 10:51:53 AM	rw-r--r--	n.yengalareddy
fastqs		6/24/2026 11:54:55 AM	rw-r--r--	n.yengalareddy
licenses		6/24/2026 10:51:54 AM	rw-r--r--	n.yengalareddy
modules		6/24/2026 10:51:54 AM	rw-r--r--	n.yengalareddy
output-202606241...		6/24/2026 1:25:01 PM	rw-r--r--	n.yengalareddy
work		6/24/2026 1:25:00 PM	rw-r--r--	n.yengalareddy
CHANGELOG.md	12 KB	6/24/2026 10:51:53 AM	rw-r--r--	n.yengalareddy
environment.yaml	3 KB	6/24/2026 10:51:53 AM	rw-r--r--	n.yengalareddy
LICENSE	12 KB	6/24/2026 10:51:53 AM	rw-r--r--	n.yengalareddy
nextflow.config	6 KB	6/24/2026 10:51:54 AM	rw-r--r--	n.yengalareddy
params.yaml	1 KB	6/24/2026 12:06:58 PM	rw-r--r--	n.yengalareddy
README.md	10 KB	6/24/2026 10:51:53 AM	rw-r--r--	n.yengalareddy
sanibel.35647270.err	1 KB	6/24/2026 12:22:40 PM	rw-r--r--	n.yengalareddy
sanibel.35647270.out	441 KB	6/24/2026 1:25:01 PM	rw-r--r--	n.yengalareddy
sanibel.nf	9 KB	6/24/2026 10:51:54 AM	rw-r--r--	n.yengalareddy
sanibel.sh	1 KB	6/24/2026 12:21:06 PM	rw-r--r--	n.yengalareddy



APPLICATION CONT.

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/Sanibel0727/output-20260624162501/

Name	Size	Changed	Rights	Owner
..		6/24/2026 1:25:01 PM	rw-r--r--	n.yengalareddy
J4-FL-M08815-240...		6/24/2026 1:22:02 PM	rw-r--r--	n.yengalareddy
J7-FL-M08815-240...		6/24/2026 1:20:28 PM	rw-r--r--	n.yengalareddy
J8-FL-M08815-240...		6/24/2026 1:19:01 PM	rw-r--r--	n.yengalareddy
J9-FL-M08815-240...		6/24/2026 1:19:59 PM	rw-r--r--	n.yengalareddy
J17-FL-M08815-24...		6/24/2026 1:21:13 PM	rw-r--r--	n.yengalareddy
J36-FL-M08815-24...		6/24/2026 1:21:57 PM	rw-r--r--	n.yengalareddy
J40-FL-M08815-24...		6/24/2026 1:20:07 PM	rw-r--r--	n.yengalareddy
J52-FL-M08815-24...		6/24/2026 1:19:26 PM	rw-r--r--	n.yengalareddy
J100-FL-M08815-2...		6/24/2026 1:19:45 PM	rw-r--r--	n.yengalareddy
J102-FL-M08815-2...		6/24/2026 1:20:18 PM	rw-r--r--	n.yengalareddy
J120-FL-M08815-2...		6/24/2026 1:20:30 PM	rw-r--r--	n.yengalareddy
J217-FL-M08815-2...		6/24/2026 1:19:28 PM	rw-r--r--	n.yengalareddy
J414-FL-M08815-2...		6/24/2026 1:20:42 PM	rw-r--r--	n.yengalareddy
J756-FL-M08815-2...		6/24/2026 1:20:52 PM	rw-r--r--	n.yengalareddy
J762-FL-M08815-2...		6/24/2026 1:20:38 PM	rw-r--r--	n.yengalareddy
J780-FL-M08815-2...		6/24/2026 1:20:49 PM	rw-r--r--	n.yengalareddy
sum_report.txt	4 KB	6/24/2026 1:25:01 PM	rw-r--r--	n.yengalareddy

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/Sanibel0727/

Name	Size	Changed	Rights	Owner
..		6/24/2026 12:05:38 PM	rw-r--r--	n.yengalareddy
assets		6/24/2026 10:51:53 AM	rw-r--r--	n.yengalareddy
bin		6/24/2026 10:51:53 AM	rw-r--r--	n.yengalareddy
db		6/24/2026 10:51:53 AM	rw-r--r--	n.yengalareddy
fastqs		6/24/2026 11:54:55 AM	rw-r--r--	n.yengalareddy
licenses		6/24/2026 10:51:54 AM	rw-r--r--	n.yengalareddy
modules		6/24/2026 10:51:54 AM	rw-r--r--	n.yengalareddy
output-202606241...		6/24/2026 1:25:01 PM	rw-r--r--	n.yengalareddy
work		6/24/2026 1:25:00 PM	rw-r--r--	n.yengalareddy
CHANGELOG.md	12 KB	6/24/2026 10:51:53 AM	rw-r--r--	n.yengalareddy
environment.yaml	3 KB	6/24/2026 10:51:53 AM	rw-r--r--	n.yengalareddy
LICENSE	12 KB	6/24/2026 10:51:53 AM	rw-r--r--	n.yengalareddy
nextflow.config	6 KB	6/24/2026 10:51:54 AM	rw-r--r--	n.yengalareddy
params.yaml	1 KB	6/24/2026 12:06:58 PM	rw-r--r--	n.yengalareddy
README.md	10 KB	6/24/2026 10:51:53 AM	rw-r--r--	n.yengalareddy
sanibel.35647270.err	1 KB	6/24/2026 12:22:40 PM	rw-r--r--	n.yengalareddy
sanibel.35647270.out	441 KB	6/24/2026 1:25:01 PM	rw-r--r--	n.yengalareddy
sanibel.nf	9 KB	6/24/2026 10:51:54 AM	rw-r--r--	n.yengalareddy
sanibel.sh	1 KB	6/24/2026 12:21:06 PM	rw-r--r--	n.yengalareddy

`sbatch sanibel.sh`

APPLICATION CONT.

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/Sanibel0727/output-20260624162501/

Name	Size	Changed	Rights	Owner
..		6/24/2026 1:25:01 PM	rw-r--r--	n.yengalareddy
J4-FL-M08815-240...		6/24/2026 1:22:02 PM	rw-r--r--	n.yengalareddy
J7-FL-M08815-240...		6/24/2026 1:20:28 PM	rw-r--r--	n.yengalareddy
J8-FL-M08815-240...		6/24/2026 1:19:01 PM	rw-r--r--	n.yengalareddy
J9-FL-M08815-240...		6/24/2026 1:19:59 PM	rw-r--r--	n.yengalareddy
J17-FL-M08815-24...		6/24/2026 1:21:13 PM	rw-r--r--	n.yengalareddy
J36-FL-M08815-24...		6/24/2026 1:21:57 PM	rw-r--r--	n.yengalareddy
J40-FL-M08815-24...		6/24/2026 1:20:07 PM	rw-r--r--	n.yengalareddy
J52-FL-M08815-24...		6/24/2026 1:19:26 PM	rw-r--r--	n.yengalareddy
J100-FL-M08815-2...		6/24/2026 1:19:45 PM	rw-r--r--	n.yengalareddy
J102-FL-M08815-2...		6/24/2026 1:20:18 PM	rw-r--r--	n.yengalareddy
J120-FL-M08815-2...		6/24/2026 1:20:30 PM	rw-r--r--	n.yengalareddy
J217-FL-M08815-2...		6/24/2026 1:19:28 PM	rw-r--r--	n.yengalareddy
J414-FL-M08815-2...		6/24/2026 1:20:42 PM	rw-r--r--	n.yengalareddy
J756-FL-M08815-2...		6/24/2026 1:20:52 PM	rw-r--r--	n.yengalareddy
J762-FL-M08815-2...		6/24/2026 1:20:38 PM	rw-r--r--	n.yengalareddy
J780-FL-M08815-2...		6/24/2026 1:20:49 PM	rw-r--r--	n.yengalareddy
sum_report.txt	4 KB	6/24/2026 1:25:01 PM	rw-r--r--	n.yengalareddy

sampleID	speciesID_mash	nearest_neighbor_mash	mash_distance	speciesID_kraken	kraken_percent	mlst_scheme	mlst_st	serotype	num_clean_reads	avg_readlength	avg_read_qual	est_coverage	num_contigs	longest_contig	N50	L50	total_length	gc_content	annotated_cds
J100-FL-M08815-240126	Escherichia_coli	GCF_000692815.1	0.00492718	Escherichia coli	49.61	ecoli_achtman_4	410	O8:H9	1865832	226.99	37.09	87.17	82	339507	203075	10	4858486	50.55	4533
J102-FL-M08815-240126	Klebsiella_pneumoniae	NZ_CCGO	0.00308338	Klebsiella pneumoniae	81.63	ecoli_achtman_4	11307	ST307	1810976	219.9	36.79	69.52	66	526876	355131	7	5728073	56.82	5305
J120-FL-M08815-240126	Pseudomonas_aeruginosa	GCF_000629065.1	0.0037263	Pseudomonas aeruginosa	91.77	paeruginosa	244	O2	1917736	211.78	36.36	55.76	115	685085	173230	12	7283745	65.43	6797
J17-FL-M08815-240126	Escherichia_coli	NZ_JHHE	0.00337267	Escherichia coli	56.62	ecoli_achtman_4	17	O103:H2	2119802	226.71	37.06	92.3	213	348504	132046	13	5206431	50.45	4967
J217-FL-M08815-240126	Acinetobacter_baumannii	NZ_AYGJ	0.00393593	Acinetobacter baumannii	76.98	abaumannii_2	2	KL23(unknown (KL23))/OCL1(OC1)	1590920	226.35	37.74	93.43	66	253267	160683	10	3854162	39.02	3602
J36-FL-M08815-240126	Klebsiella_pneumoniae	GCF_000445405.1	0.00508714	Klebsiella pneumoniae	84.1	ecoli_achtman_4	14464	ST11	2181456	218.57	36.64	88.21	62	752655	203322	8	5405432	57.52	5042
J4-FL-M08815-240126	Escherichia_coli	NZ_JNRV	0.00936594	Escherichia coli	40.47	ecoli_achtman_4	10	O29:H4	3006450	227.56	37.18	131.69	168	267838	101672	16	5195327	50.73	4930
J40-FL-M08815-240126	Enterobacter_sp.	GCF_000958405.1	0.0111761	Enterobacter hormaechei	18.4	ecloacae	102	No data	1347996	219.73	36.63	61.05	75	326759	160256	11	4851067	55.13	4540
J414-FL-M08815-240126	Escherichia_coli	NZ_AVRE	0.00243596	Escherichia coli	66.83	ecoli_achtman_4	11	O157:H7	2575052	226.73	37.08	109.46	159	374706	150818	11	5333765	50.23	5068
J52-FL-M08815-240126	Escherichia_coli	GCF_000461215.1	0.00680395	Escherichia coli	39.07	ecoli_achtman_4	10	O54:H2	2645448	225.13	37.12	120.86	110	363200	119254	13	4927766	50.47	4690
J7-FL-M08815-240126	Escherichia_coli	NZ_JHGK	0.00243596	Escherichia coli	57.21	ecoli_achtman_4	21	O26:H11	1893390	226.08	37.11	81.92	194	314867	126353	15	5224808	50.4	5016
J756-FL-M08815-240126	Salmonella_enterica	GCF_000171495.1	0.00331442	Salmonella enterica	96.21	salmonella	365	3,10:r:z6(Weltevreden)	1530610	221.32	36.85	68.5	73	332968	162252	10	4945396	52.04	4625
J762-FL-M08815-240126	Salmonella_enterica	NZ_AJMO	0.00518381	Salmonella enterica	96.83	salmonella	118	8:e,h:1,2(Newport)	1144434	225.27	36.75	53.98	41	959478	429968	4	4775969	52.19	4474
J780-FL-M08815-240126	Salmonella_enterica	GCF_000493295.1	0.00737682	Salmonella enterica	93.74	ecoli_achtman_4	9187	13:z:1,6(Poona)	1428284	221.39	36.72	63.88	82	602312	341732	6	4949946	52.04	4721
J8-FL-M08815-240126	Escherichia_coli	NZ_AIBR	0.00638544	Escherichia coli	53	ecoli_achtman_4	29	O118/O151/O151:H11	2452914	227.54	37.06	103.68	253	211567	83289	23	5382884	50.26	5220
J9-FL-M08815-240126	Escherichia_coli	GCF_000458955.1	0.00375609	Escherichia coli	50.87	ecoli_achtman_4	1248	NT:H21	2227612	226.32	37.04	108	48	740618	237932	7	4668075	50.71	4363

APPLICATION CONT.

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/Sanibel0727/output-20260624162501/J4-FL-M08815-240126/

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APPLICATION CONT.

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bbtools		6/24/2026 12:25:22 PM	rw-r--r--	n.yengalareddy	
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trimmomatic		6/24/2026 12:24:04 PM	rw-r--r--	n.yengalareddy	

Protein id	Contig id	Start	Stop	Strand	Element symbol	Element name	Scope	Type	Subtype	Class	Subclass	Method	Target length	Reference sequence length	% Coverage of reference	% Identity to reference	Alignment length	Closest reference accession	Closest reference name	HMM accession	HMM desc	
NA	3	1E+05	1E+05	+	sigA	serine protease autotransporter toxin SigA	plus	VIOLENCE	VIOLENCE	NA	NA	BLASTX	1285	1285	100	97.67	1285	AA044449.1	serine protease autotransporter toxin SigA	NA	NA	
NA	3	2E+05	2E+05	+	sigA	serine protease autotransporter toxin SigA	plus	VIOLENCE	VIOLENCE	NA	NA	BLASTX	1285	1285	100	99.92	1285	AA044449.1	serine protease autotransporter toxin SigA	NA	NA	
NA	4	2E+05	2E+05	+	terD	tellurium resistance membrane protein TerD	plus	STRESS	METAL	TELLURIUM	TELLURIUM	BLASTX	192	192	100	95.31	192	AA068292.1	tellurium resistance membrane protein TerD	NA	NA	
NA	4	2E+05	2E+05	+	terZ	tellurium resistance-associated protein TerZ	plus	STRESS	METAL	TELLURIUM	TELLURIUM	BLASTX	193	193	100	96.36	193	AA068292.1	tellurium resistance-associated protein TerZ	NA	NA	
NA	4	2E+05	2E+05	+	terW	tellurium resistance protein TerW	plus	STRESS	METAL	TELLURIUM	TELLURIUM	BLASTX	155	155	100	98.71	155	AA044736.1	tellurium resistance protein TerW	NA	NA	
NA	5	1E+05	1E+05	+	blaEC	BlaEC family class C beta-lactamase	plus	AMR	AMR	BETA-LACTAM	BETA-LACTAM	EXACTX	377	377	100	100	377	AA077110.1	BlaEC family class C beta-lactamase	NA	NA	
NA	6	2E+05	2E+05	+	asr	acid resistance repetitive basic protein Asr	plus	STRESS	ACID	NA	NA	EXACTX	102	102	100	100	102	BA015331.2	acid resistance repetitive basic protein Asr	NA	NA	
NA	9	1E+05	1E+05	+	espX1	type III secretion system effector EspX1	plus	VIOLENCE	VIOLENCE	NA	NA	BLASTX	473	473	100	89.01	473	AD054679.1	type III secretion system effector EspX1	NA	NA	
NA	15	43052	43951	+	fieF	CDF family cation-efflux transporter FieF	plus	STRESS	METAL	NA	NA	EXACTX	300	300	100	100	300	BA07395.1	CDF family cation-efflux transporter FieF	NA	NA	
NA	16	5767	7586	+	ynfQ	yersiniabactin ABC transporter ATP-binding/permease protein YnfQ	plus	VIOLENCE	VIOLENCE	NA	NA	BLASTX	600	600	100	99.83	600	AA039584.1	yersiniabactin ABC transporter ATP-binding/permease protein YnfQ	NA	NA	
NA	16	7576	9375	+	ynfP	yersiniabactin ABC transporter ATP-binding/permease protein YnfP	plus	VIOLENCE	VIOLENCE	NA	NA	BLASTX	600	600	100	99.83	600	CAA21388.1	yersiniabactin ABC transporter ATP-binding/permease protein YnfP	NA	NA	
NA	17	87097	88278	+	emrD	multidrug efflux MFS transporter EmrD	plus	AMR	AMR	EFFLUX	EFFLUX	EXACTX	394	394	100	100	394	BA077620.1	multidrug efflux MFS transporter EmrD	NA	NA	
NA	20	17661	19706	+	ireA	TonB-dependent siderophore receptor IreA	plus	VIOLENCE	VIOLENCE	NA	NA	BLASTX	682	682	100	99.56	682	AA083596.1	TonB-dependent siderophore receptor IreA	NA	NA	
NA	20	84061	84873	+	su2	sulfonamide-resistant dihydropteroate synthase Su2	core	AMR	AMR	SULFONAMIDE	SULFONAMIDE	EXACTX	271	271	100	100	271	WP_001043260.1	sulfonamide-resistant dihydropteroate synthase Su2	NA	NA	
NA	20	87452	88325	+	aph(III)-Id	aminoglycoside O-phosphotransferase APH(III)-Id	core	AMR	AMR	AMINOGLYCOSIDE	STREPTOMYCIN	EXACTX	278	278	100	99.63	278	WP_000409968.1	aminoglycoside O-phosphotransferase APH(III)-Id	NA	NA	
NA	20	86328	89128	+	aph(3')-Ib	aminoglycoside O-phosphotransferase APH(3')-Ib	core	AMR	AMR	AMINOGLYCOSIDE	STREPTOMYCIN	BLASTX	267	267	100	99.63	267	WP_001082319.1	aminoglycoside O-phosphotransferase APH(3')-Ib	NA	NA	
NA	26	40541	41770	+	mdtM	multidrug efflux MFS transporter MdtM	plus	AMR	AMR	EFFLUX	EFFLUX	BLASTX	410	410	100	97.8	410	AA077293.1	multidrug efflux MFS transporter MdtM	NA	NA	
NA	31	2	793	+	siIC	Cu(+)/Ag(+)-efflux RND transporter outer membrane channel SiIC	plus	STRESS	METAL	COPPER/SILVER	COPPER/SILVER	PARTIAL_CONFIG_ENDX	264	461	57.27	96.59	264	SPD96884.1	Cu(+)/Ag(+)-efflux RND transporter outer membrane channel SiIC	NA	NA	
NA	31	963	1660	+	siIR	copper/silver response regulator transcription factor SiIR	plus	STRESS	METAL	COPPER/SILVER	COPPER/SILVER	BLASTX	226	226	100	97.79	226	SPD96883.1	copper/silver response regulator transcription factor SiIR	NA	NA	
NA	31	1656	3128	+	sis	copper/silver sensor histidine kinase Sis	plus	STRESS	METAL	COPPER/SILVER	COPPER/SILVER	BLASTX	491	491	100	98.57	491	SPD96882.1	copper/silver sensor histidine kinase Sis	NA	NA	
NA	31	382	3810	+	siE	silver-binding protein SiE	plus	STRESS	METAL	SILVER	SILVER	BLASTX	143	143	100	99.3	143	AA011743.1	silver-binding protein SiE	NA	NA	
NA	31	40526	40676	+	arsR	As(III)-sensing metalloregulatory transcriptional repressor ArsR	plus	STRESS	METAL	ARSENIC	ARSENIC	EXACTX	117	117	100	100	117	BA07793.1	As(III)-sensing metalloregulatory transcriptional repressor ArsR	NA	NA	
NA	31	42235	42657	+	arsC	glutaredoxin-dependent arsenate reductase	plus	STRESS	METAL	ARSENIC	ARSENATE	BLASTX	141	141	100	94.33	141	AA021096.1	glutaredoxin-dependent arsenate reductase	NA	NA	
NA	33	5529	8630	+	acrF	multidrug efflux RND transporter permease subunit AcrF	plus	AMR	AMR	EFFLUX	EFFLUX	EXACTX	1034	1034	100	100	1034	AA076298.1	multidrug efflux RND transporter permease subunit AcrF	NA	NA	
NA	40	23063	23392	+	emrE	multidrug efflux SMR transporter EmrE	plus	STRESS	BIODIE	EFFLUX	EFFLUX	BLASTX	110	110	100	98.18	110	CAA77936.1	multidrug efflux SMR transporter EmrE	NA	NA	
NA	45	8502	12749	+	fdeC	inverse autotransporter adhesin FdeC	plus	VIOLENCE	VIOLENCE	NA	NA	BLASTX	1416	1416	100	94.14	1416	AB058222.1	inverse autotransporter adhesin FdeC	NA	NA	
NA	57	9194	11293	+	mchF	microcin H47 export transporter peptidase/ATP-binding subunit MchF	plus	VIOLENCE	VIOLENCE	NA	NA	EXACTX	700	700	100	100	700	AF058565.1	microcin H47 export transporter peptidase/ATP-binding subunit MchF	NA	NA	
NA	57	15016	15040	+	mchB	microcin H47	plus	EXACTX	75	75	75	75	EXACTX	75	75	100	100	75	CB854534.2	microcin H47	NA	NA
NA	59	13586	15307	+	lucA	aerobactin synthase LucA	plus	VIOLENCE	VIOLENCE	NA	NA	BLASTX	574	576	100	98.44	576	EG025746.1	aerobactin synthase LucA	NA	NA	
NA	59	15311	16255	+	lucB	Ni(II)-hydroxylase O-acetyltransferase LucB	plus	VIOLENCE	VIOLENCE	NA	NA	BLASTX	315	315	100	99.37	315	AA082074.1	Ni(II)-hydroxylase O-acetyltransferase LucB	NA	NA	
NA	59	16258	17997	+	lucC	NiS family aerobactin synthetase LucC	plus	VIOLENCE	VIOLENCE	NA	NA	BLASTX	580	580	100	99.48	580	AA082073.1	NiS family aerobactin synthetase LucC	NA	NA	
NA	59	17997	19331	+	lucD	NADPH-dependent L-lysine Ni(II)-monooxygenase LucD	plus	VIOLENCE	VIOLENCE	NA	NA	BLASTX	445	445	100	99.55	445	AA044749.1	NADPH-dependent L-lysine Ni(II)-monooxygenase LucD	NA	NA	
NA	59	19340	21532	+	lutA	ferric aerobactin receptor LutA	plus	VIOLENCE	VIOLENCE	NA	NA	BLASTX	731	731	100	99.59	731	AA045165.2	ferric aerobactin receptor LutA	NA	NA	
NA	62	70	972	+	mph(A)	Mph(A) family macrolide 2'-phosphotransferase	core	AMR	AMR	MACROLIDE	AZITHROMYCIN/ERYTHROMYCIN/SPRAMYCIN/TELITHROMYCIN	EXACTX	301	301	100	100	301	WP_000215391.1	Mph(A) family macrolide 2'-phosphotransferase	NA	NA	
NA	62	972	2207	+	mrx(A)	phosphorylated macrolide efflux MFS transporter Mrx(A)	core	AMR	AMR	MACROLIDE	AZITHROMYCIN/ERYTHROMYCIN/SPRAMYCIN/TELITHROMYCIN	EXACTX	412	412	100	100	412	WP_000004159.1	phosphorylated macrolide efflux MFS transporter Mrx(A)	NA	NA	
NA	62	6289	7125	+	suI	sulfonamide-resistant dihydropteroate synthase SuI	plus	AMR	AMR	SULFONAMIDE	SULFONAMIDE	EXACTX	279	279	100	97.79	279	WP_000259031.1	sulfonamide-resistant dihydropteroate synthase SuI	NA	NA	
NA	62	7122	7466	+	qacDelta1	quaternary ammonium compound efflux SMR transporter QacE delta 1	core	STRESS	BIODIE	QUATERNARY AMMONIUM	QUATERNARY AMMONIUM	ALLELEX	115	115	100	100	115	WP_000679427.1	quaternary ammonium compound efflux SMR transporter QacE delta 1	NA	NA	
NA	62	7675	8460	+	aadA5	ANT(3'')-Ia family aminoglycoside nucleotidyltransferase AadA5	core	AMR	AMR	AMINOGLYCOSIDE	STREPTOMYCIN	EXACTX	262	262	100	100	262	WP_000503573.1	ANT(3'')-Ia family aminoglycoside nucleotidyltransferase AadA5	NA	NA	
NA	62	8594	9064	+	dfrA17	trimethoprim-resistant dihydrofolate reductase DfrA17	core	AMR	AMR	TRIMETHOPRIM	TRIMETHOPRIM	EXACTX	157	157	100	100	157	WP_001389366.1	trimethoprim-resistant dihydrofolate reductase DfrA17	NA	NA	
NA	62	15546	16742	+	tet(A)	tetracycline efflux MFS transporter Tet(A)	core	AMR	AMR	TETRACYCLINE	TETRACYCLINE	EXACTX	399	399	100	100	399	WP_000804064.1	tetracycline efflux MFS transporter Tet(A)	NA	NA	
NA	65	627	977	+	siF	Cu(+)/Ag(+)-efflux RND transporter periplasmic metallochaperone SiF	plus	STRESS	METAL	COPPER/SILVER	COPPER/SILVER	BLASTX	117	117	100	97.44	117	K017342.1	Cu(+)/Ag(+)-efflux RND transporter periplasmic metallochaperone SiF	NA	NA	
NA	65	1094	2383	+	siB	Cu(+)/Ag(+)-efflux RND transporter periplasmic adaptor subunit SiB	plus	STRESS	METAL	COPPER/SILVER	COPPER/SILVER	BLASTX	430	430	100	96.28	430	AA031748.1	Cu(+)/Ag(+)-efflux RND transporter periplasmic adaptor subunit SiB	NA	NA	
NA	65	2397	5540	+	siA	Cu(+)/Ag(+)-efflux RND transporter permease subunit SiA	plus	STRESS	METAL	COPPER/SILVER	COPPER/SILVER	BLASTX	1048	1048	100	98.66	1048	AA011749.1	Cu(+)/Ag(+)-efflux RND transporter permease subunit SiA	NA	NA	
NA	65	6176	8641	+	siP	Ag(+)-translocating P-type ATPase SiP	plus	STRESS	METAL	SILVER	SILVER	BLASTX	822	824	99.64	95.14	823	AA011750.1	Ag(+)-translocating P-type ATPase SiP	NA	NA	
NA	65	10626	12440	+	pcoA	multicopper oxidase PcoA	plus	STRESS	METAL	COPPER	COPPER	BLASTX	605	605	100	99.5	605	CAA58525.1	multicopper oxidase PcoA	NA	NA	
NA	65	12449	13336	+	pcoB	copper-binding protein PcoB	plus	STRESS	METAL	COPPER	COPPER	EXACTX	296	296	100	100	296	CAA58526.1	copper-binding protein PcoB	NA	NA	
NA	65	13379	13756	+	pcoC	copper resistance system metallochaperone PcoC	plus	STRESS	METAL	COPPER	COPPER	EXACTX	126	126	100	100	126	CAA58527.1	copper resistance system metallochaperone PcoC	NA	NA	
NA	65	13764	14690	+	pcoD	copper resistance inner membrane protein PcoD	plus	STRESS	METAL	COPPER	COPPER	EXACTX	309	309	100	100	309	CAA58528.1	copper resistance inner membrane protein PcoD	NA	NA	
NA	65	14748	15425	+	pcoR	copper response regulator transcription factor PcoR	plus	STRESS	METAL	COPPER	COPPER	EXACTX	226	226	100	100	226	CAA58529.1	copper response regulator transcription factor PcoR	NA	NA	
NA	88	2	1063	+	pcoS	copper resistance membrane spanning protein PcoS	plus	STRESS	METAL	COPPER	COPPER	PARTIAL_CONFIG_ENDX	354	426	75.97	99.72	354	CAA58530.1	copper resistance membrane spanning protein PcoS	NA	NA	
NA	88	1284	1715	+	pcoE	copper resistance system metallochaperone PcoE	plus	STRESS	METAL	COPPER	COPPER	BLASTX	144	144	100	92.36	144	CAA58532.1	copper resistance system metallochaperone PcoE	NA	NA	
NA	107	444	1583	+	iha	bifunctional siderophore receptor/adhesin Iha	plus	VIOLENCE	VIOLENCE	NA	NA	PARTIAL_CONFIG_ENDX	300	696	54.6	100	300	AA055323.1	bifunctional siderophore receptor/adhesin Iha	NA	NA	

APPLICATION CONT.

/.../bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/Sanibel0727/output-20260624162501/J4-FL-M08815-240126/assembly/					
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prokka		6/24/2026 1:23:50 PM	rw-r--r--	n.yengalareddy	
quast_results		6/24/2026 1:20:39 PM	rw-r--r--	n.yengalareddy	
J4-FL-M08815-240...	5,17...	6/24/2026 1:18:19 PM	rw-r--r--	n.yengalareddy	

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/Sanibel0727/output-20260624162501/J4-FL-M08815-240126/					
Name	Size	Changed	Rights	Owner	
..		6/24/2026 1:25:01 PM	rw-r--r--	n.yengalareddy	
amrfinder		6/24/2026 1:21:03 PM	rw-r--r--	n.yengalareddy	
assembly		6/24/2026 1:23:50 PM	rw-r--r--	n.yengalareddy	
bbtools		6/24/2026 12:25:22 PM	rw-r--r--	n.yengalareddy	
fastqc		6/24/2026 12:24:18 PM	rw-r--r--	n.yengalareddy	
fastqc2		6/24/2026 12:25:42 PM	rw-r--r--	n.yengalareddy	
kraken_out		6/24/2026 12:26:13 PM	rw-r--r--	n.yengalareddy	
mash_output		6/24/2026 12:26:00 PM	rw-r--r--	n.yengalareddy	
mlst		6/24/2026 1:20:57 PM	rw-r--r--	n.yengalareddy	
multiqc		6/24/2026 1:09:47 PM	rw-r--r--	n.yengalareddy	
plasmid		6/24/2026 1:22:02 PM	rw-r--r--	n.yengalareddy	
serotypefinder		6/24/2026 1:21:22 PM	rw-r--r--	n.yengalareddy	
trimmomatic		6/24/2026 12:24:04 PM	rw-r--r--	n.yengalareddy	

Assembly	J4-FL-M08815-240126
# contigs (>= 0 bp)	215
# contigs (>= 1000 bp)	125
# contigs (>= 5000 bp)	84
# contigs (>= 10000 bp)	70
# contigs (>= 25000 bp)	54
# contigs (>= 50000 bp)	37
Total length (>= 0 bp)	5213555
Total length (>= 1000 bp)	5163141
Total length (>= 5000 bp)	5077451
Total length (>= 10000 bp)	4964639
Total length (>= 25000 bp)	4678098
Total length (>= 50000 bp)	4069136
# contigs	168
Largest contig	267838
Total length	5195327
GC (%)	50.73
N50	101672
N90	25534
auN	116794.9
L50	16
L90	54
# N's per 100 kbp	0

APPLICATION CONT.

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/Sanibel0727/output-20260624162501/J4-FL-M08815-240126/bbtools/					
Name	Size	Changed	Rights	Owner	
..		6/24/2026 1:22:02 PM	rw-r--r--	n.yengalareddy	
J4-FL-M08815-240...	1 KB	6/24/2026 12:24:41 PM	rw-r--r--	n.yengalareddy	
J4-FL-M08815-240...	1 KB	6/24/2026 12:25:22 PM	rw-r--r--	n.yengalareddy	
J4-FL-M08815-240...	216,...	6/24/2026 12:24:42 PM	rw-r--r--	n.yengalareddy	
J4-FL-M08815-240...	216,...	6/24/2026 12:25:23 PM	rw-r--r--	n.yengalareddy	
J4-FL-M08815-240...	226,...	6/24/2026 12:24:42 PM	rw-r--r--	n.yengalareddy	
J4-FL-M08815-240...	226,...	6/24/2026 12:25:23 PM	rw-r--r--	n.yengalareddy	

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/Sanibel0727/output-20260624162501/J4-FL-M08815-240126/					
Name	Size	Changed	Rights	Owner	
..		6/24/2026 1:25:01 PM	rw-r--r--	n.yengalareddy	
amrfinder		6/24/2026 1:21:03 PM	rw-r--r--	n.yengalareddy	
assembly		6/24/2026 1:23:50 PM	rw-r--r--	n.yengalareddy	
bbtools		6/24/2026 12:25:22 PM	rw-r--r--	n.yengalareddy	
fastqc		6/24/2026 12:24:18 PM	rw-r--r--	n.yengalareddy	
fastqc2		6/24/2026 12:25:42 PM	rw-r--r--	n.yengalareddy	
kraken_out		6/24/2026 12:26:13 PM	rw-r--r--	n.yengalareddy	
mash_output		6/24/2026 12:26:00 PM	rw-r--r--	n.yengalareddy	
mlst		6/24/2026 1:20:57 PM	rw-r--r--	n.yengalareddy	
multiqc		6/24/2026 1:09:47 PM	rw-r--r--	n.yengalareddy	
plasmid		6/24/2026 1:22:02 PM	rw-r--r--	n.yengalareddy	
serotypefinder		6/24/2026 1:21:22 PM	rw-r--r--	n.yengalareddy	
trimmomatic		6/24/2026 12:24:04 PM	rw-r--r--	n.yengalareddy	

#File	J4-FL-M08815-240126_trim_1.fastq.gz	J4-FL-M08815-240126_trim_2.fastq.gz
#Total	3006452	
#Matched	1483	0.05%
#Name	Reads	ReadsPct
PhiX_read2_adapter	192	0.01%
I5_Adapter_Nextera	196	0.01%
I5_Primer_Nextera_XT_and_Nextera_Enrichment_[N/S/E]501	118	0.00%
I7_Nextera_Transposase_2	114	0.00%
I5_Nextera_Transposase_1	107	0.00%
Reverse_adapter	99	0.00%
Bisulfite_R1	84	0.00%
PhiX_read1_adapter	78	0.00%
TruSeq_Universal_Adapter	77	0.00%
RNA_PCR_Primer_Index_1_(RP1)_2,9	59	0.00%
RNA_Adapter_(RA5)_part_#_15013205	62	0.00%
Nextera_LMP_Read2_External_Adapter	50	0.00%
Nextera_LMP_Read1_External_Adapter	46	0.00%
I7_Adapter_Nextera_No_Barcode	45	0.00%
TruSeq_Adapter_Index_1_6	36	0.00%
Illumina Small RNA v1.5 3p Adapter	35	0.00%
I5_Nextera_Transposase_2	28	0.00%
I7_Nextera_Transposase_1	20	0.00%
I7_Primer_Nextera_XT_and_Nextera_Enrichment_N701	16	0.00%
RNA_PCR_Primer_(RP1)_part_#_15013198	9	0.00%
Bisulfite_R2	9	0.00%
TruSeq_Adapter_Index_14	3	0.00%

APPLICATION CONT.

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/Sanibel0727/output-20260624162501/J4-FL-M08815-240126/fastqc/					
Name	Size	Changed	Rights	Owner	
..		6/24/2026 1:22:02 PM	rw-r--r--	n.yengalareddy	
J4-FL-M08815-240...	666 KB	6/24/2026 12:24:18 PM	rw-r--r--	n.yengalareddy	
J4-FL-M08815-240...	386 KB	6/24/2026 12:24:18 PM	rw-r--r--	n.yengalareddy	
J4-FL-M08815-240...	676 KB	6/24/2026 12:24:18 PM	rw-r--r--	n.yengalareddy	
J4-FL-M08815-240...	407 KB	6/24/2026 12:24:18 PM	rw-r--r--	n.yengalareddy	

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/Sanibel0727/output-20260624162501/J4-FL-M08815-240126/					
Name	Size	Changed	Rights	Owner	
..		6/24/2026 1:25:01 PM	rw-r--r--	n.yengalareddy	
amrfinder		6/24/2026 1:21:03 PM	rw-r--r--	n.yengalareddy	
assembly		6/24/2026 1:23:50 PM	rw-r--r--	n.yengalareddy	
bbtools		6/24/2026 12:25:22 PM	rw-r--r--	n.yengalareddy	
fastqc		6/24/2026 12:24:18 PM	rw-r--r--	n.yengalareddy	
fastqc2		6/24/2026 12:25:42 PM	rw-r--r--	n.yengalareddy	
kraken_out		6/24/2026 12:26:13 PM	rw-r--r--	n.yengalareddy	
mash_output		6/24/2026 12:26:00 PM	rw-r--r--	n.yengalareddy	
mlst		6/24/2026 1:20:57 PM	rw-r--r--	n.yengalareddy	
multiqc		6/24/2026 1:09:47 PM	rw-r--r--	n.yengalareddy	
plasmid		6/24/2026 1:22:02 PM	rw-r--r--	n.yengalareddy	
serotypefinder		6/24/2026 1:21:22 PM	rw-r--r--	n.yengalareddy	
trimmomatic		6/24/2026 12:24:04 PM	rw-r--r--	n.yengalareddy	

FastQC Report

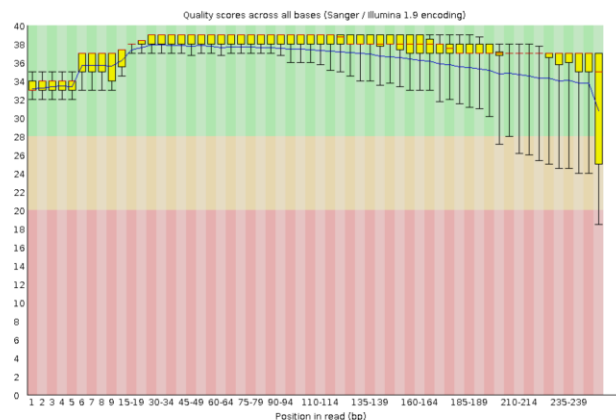
Summary

- Basic Statistics
- Per base sequence quality
- Per tile sequence quality
- Per sequence quality scores
- Per base sequence content
- Per sequence GC content
- Per base N content
- Sequence Length Distribution
- Sequence Duplication Levels
- Overrepresented sequences
- Adapter Content

Basic Statistics

Measure	Value
Filename	J4-FL-M08815-240126_S2_L001_R1_001.fastq.gz
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	1699717
Total Bases	407.2 Mbp
Sequences flagged as poor quality	0
Sequence length	35-251
%GC	50

Per base sequence quality



APPLICATION CONT.

/.../n.yengalareddy/repos/bphl-molecular/analysis/Sanibel0727/output-20260624162501/J4-FL-M08815-240126/kraken_out/

Name	Size	Changed	Rights	Owner
..		6/24/2026 1:22:02 PM	rw-r--r--	n.yengalareddy
J4-FL-M08815-240...	36 KB	6/24/2026 12:26:13 PM	rw-r--r--	n.yengalareddy

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/Sanibel0727/output-20260624162501/J4-FL-M08815-240126/

Name	Size	Changed	Rights	Owner
..		6/24/2026 1:25:01 PM	rw-r--r--	n.yengalareddy
amrfinder		6/24/2026 1:21:03 PM	rw-r--r--	n.yengalareddy
assembly		6/24/2026 1:23:50 PM	rw-r--r--	n.yengalareddy
bbtools		6/24/2026 12:25:22 PM	rw-r--r--	n.yengalareddy
fastqc		6/24/2026 12:24:18 PM	rw-r--r--	n.yengalareddy
fastqc2		6/24/2026 12:25:42 PM	rw-r--r--	n.yengalareddy
kraken_out		6/24/2026 12:26:13 PM	rw-r--r--	n.yengalareddy
mash_output		6/24/2026 12:26:00 PM	rw-r--r--	n.yengalareddy
mlst		6/24/2026 1:20:57 PM	rw-r--r--	n.yengalareddy
multiqc		6/24/2026 1:09:47 PM	rw-r--r--	n.yengalareddy
plasmid		6/24/2026 1:22:02 PM	rw-r--r--	n.yengalareddy
serotypefinder		6/24/2026 1:21:22 PM	rw-r--r--	n.yengalareddy
trimmomatic		6/24/2026 12:24:04 PM	rw-r--r--	n.yengalareddy

0.94	14102	14102 U	0 unclassified	
99.06	1489123	8593 R	1 root	
98.47	1480158	5 R1	131567 cellular organisms	
98.47	1480153	5578 D	2 Bacteria	
98.09	1474528	5160 P	1224 Proteobacteria	
97.74	1469207	3192 C	1236 Gammaproteobacteria	
97.32	1462983	28406 O	91347 Enterobacterales	
95.41	1434167	779886 F	543 Enterobacteriaceae	
42.21	634484	23198 G	561 Escherichia	
40.47	608291	565377 S	562 Escherichia coli	
2.52	37857	37857 S1	585397 Escherichia coli ED1a	
0.14	2123	2123 S1	941323 Escherichia coli VR50	
0.04	636	636 S1	1382700 Escherichia coli PMV-1	
0.02	343	1 S1	244320 Escherichia coli O55:H7	
0.02	247	247 S2	701177 Escherichia coli O55:H7 str. CB9615	
0.01	95	95 S2	1048899 Escherichia coli O55:H7 str. RM12579	
0.02	262	262 S1	585055 Escherichia coli 55989	
0.01	143	143 S1	2048777 Escherichia coli O15:H11	
0.01	139	0 S1	2233553 Escherichia coli O43	
0.01	139	139 S2	1055541 Escherichia coli O43 str. RM10042	
0.01	135	135 S1	1050617 Escherichia coli UMN18	
0.01	130	130 S1	168807 Escherichia coli O127:H6	
0.01	128	128 S1	439855 Escherichia coli SMS-3-5	
0.01	120	120 S1	745156 Escherichia coli 1303	
0.01	117	117 S1	1412834 Escherichia coli FAP1	
0.01	117	117 S1	696406 Escherichia coli UMN188	
0.01	112	0 S1	861906 Escherichia coli O44:H18	
0.01	112	112 S2	216592 Escherichia coli O42	
0.01	89	89 S1	930406 Escherichia coli O157:H16	
0.01	80	80 S1	409438 Escherichia coli SE11	
0.01	77	77 S1	1329907 Escherichia coli APEC IMT5155	
0.01	77	77 S1	941280 Escherichia coli O25b:H4	
0	47	47 S1	1078034 Escherichia coli O145:H28	
0	29	20 S1	83334 Escherichia coli O157:H7	
0	6	6 S2	155864 Escherichia coli O157:H7 str. EDL933	
0	2	2 S2	1330457 Escherichia coli O157:H7 str. SS52	
0	1	1 S2	1328859 Escherichia coli O157:H7 str. SS17	
0	14	14 S1	1392658 Escherichia coli M12	
0	13	13 S1	1446746 Escherichia coli O6:H16	
0	13	6 S1	83333 Escherichia coli K-12	
0	7	6 S2	511145 Escherichia coli str. K-12 substr. MG1655	
0	1	1 S3	879462 Escherichia coli str. K-12 substr. MG1655star	
0	9	9 S1	1358422 Escherichia coli PCN061	
0	9	0 S1	1603259 Escherichia coli O139:H28	
0	9	9 S2	331111 Escherichia coli O139:H28 str. E24377A	
0	9	0 S1	498388 Escherichia coli C	
0	9	9 S2	481805 Escherichia coli ATCC 8739	
0	9	9 S1	2048776 Escherichia coli O128:H27	
0	8	8 S1	331112 Escherichia coli HS	
0	7	7 S1	362663 Escherichia coli 536	
0	6	0 S1	1072458 Escherichia coli O7:K1	
0	6	6 S2	1072459 Escherichia coli O7:K1 str. CE10	
0	5	5 S1	585034 Escherichia coli IAI1	
0	4	4 S1	1446707 Escherichia coli O25:NM	
0	3	3 S1	405955 Escherichia coli APEC O1	

APPLICATION CONT.

/.../n.yengalareddy/repos/bphl-molecular/analysis/Sanibel0727/output-20260624162501/J4-FL-M08815-240126/mash_output/					
Name	Size	Changed	Rights	Owner	
J4-FL-M08815-240...	2 KB	6/24/2026 12:26:00 PM	rw-r--r--	n.yengalareddy	

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/Sanibel0727/output-20260624162501/J4-FL-M08815-240126/					
Name	Size	Changed	Rights	Owner	
..		6/24/2026 1:25:01 PM	rw-r--r--	n.yengalareddy	
amrfinder		6/24/2026 1:21:03 PM	rw-r--r--	n.yengalareddy	
assembly		6/24/2026 1:23:50 PM	rw-r--r--	n.yengalareddy	
bbtools		6/24/2026 12:25:22 PM	rw-r--r--	n.yengalareddy	
fastqc		6/24/2026 12:24:18 PM	rw-r--r--	n.yengalareddy	
fastqc2		6/24/2026 12:25:42 PM	rw-r--r--	n.yengalareddy	
kraken_out		6/24/2026 12:26:13 PM	rw-r--r--	n.yengalareddy	
mash_output		6/24/2026 12:26:00 PM	rw-r--r--	n.yengalareddy	
mlst		6/24/2026 1:20:57 PM	rw-r--r--	n.yengalareddy	
multiqc		6/24/2026 1:09:47 PM	rw-r--r--	n.yengalareddy	
plasmid		6/24/2026 1:22:02 PM	rw-r--r--	n.yengalareddy	
serotypefinder		6/24/2026 1:21:22 PM	rw-r--r--	n.yengalareddy	
trimmomatic		6/24/2026 12:24:04 PM	rw-r--r--	n.yengalareddy	

refseq-NZ-1444083-PRJNA224116-SAMN02680241-NZ_JNRV--Escherichia_coli_3_073_06_S1_C2.fna	M08815:22:000000000-LD39V:1:1101:15965:7023	0.00936594	0	697/1000
refseq-NZ-1444269-PRJNA224116-SAMN02687475-NZ_JOQT--Escherichia_coli_5_172_05_S4_C3.fna	M08815:22:000000000-LD39V:1:1101:15965:7023	0.0097316	0	688/1000
refseq-NZ-1281242-PRJNA224116-SAMN01885954-GCF_000461255.1--Escherichia_coli_UMEA_3329_1.fna	M08815:22:000000000-LD39V:1:1101:15965:7023	0.00981378	0	686/1000
refseq-NZ-1444239-PRJNA224116-SAMN02687503-NZ_JOQN--Escherichia_coli_5_172_05_S4_C2.fna	M08815:22:000000000-LD39V:1:1101:15965:7023	0.00985499	0	685/1000
refseq-NZ-1444211-PRJNA224116-SAMN02680256-NZ_JOQP--Escherichia_coli_5_172_05_S4_C1.fna	M08815:22:000000000-LD39V:1:1101:15965:7023	0.00993767	0	683/1000
refseq-NZ-562-PRJNA224116-SAMN03075145-GCF_000948005.1--Escherichia_coli.fna	M08815:22:000000000-LD39V:1:1101:15965:7023	0.0101459	0	678/1000
refseq-NZ-1444122-PRJNA224116-SAMN02687540-NZ_JNQH--Escherichia_coli_2_210_07_S3_C1.fna	M08815:22:000000000-LD39V:1:1101:15965:7023	0.0101878	0	677/1000
refseq-NZ-754078-PRJNA179770-SAMN00015374-NZ_AFDX--Escherichia_coli_G58_1.fna	M08815:22:000000000-LD39V:1:1101:15965:7023	0.0102298	0	676/1000
refseq-NZ-1281136-PRJNA224116-SAMN01885848-GCF_000459235.1--Escherichia_coli_HVH_209_4_3062651.fna	M08815:22:000000000-LD39V:1:1101:15965:7023	0.0102718	0	675/1000
refseq-NZ-1444244-PRJNA224116-SAMN02680168-NZ_JMGW--Escherichia_coli_1_176_05_S4_C2.fna	M08815:22:000000000-LD39V:1:1101:15965:7023	0.010314	0	674/1000

APPLICATION CONT.

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/Sanibel0727/output-20260624162501/J4-FL-M08815-240126/mlst/					
Name	Size	Changed	Rights	Owner	
..		6/24/2026 1:22:02 PM	rw-r--r--	n.yengalareddy	
J4-FL-M08815-240...	1 KB	6/24/2026 1:20:57 PM	rw-r--r--	n.yengalareddy	

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/Sanibel0727/output-20260624162501/J4-FL-M08815-240126/					
Name	Size	Changed	Rights	Owner	
..		6/24/2026 1:25:01 PM	rw-r--r--	n.yengalareddy	
amrfinder		6/24/2026 1:21:03 PM	rw-r--r--	n.yengalareddy	
assembly		6/24/2026 1:23:50 PM	rw-r--r--	n.yengalareddy	
bbtools		6/24/2026 12:25:22 PM	rw-r--r--	n.yengalareddy	
fastqc		6/24/2026 12:24:18 PM	rw-r--r--	n.yengalareddy	
fastqc2		6/24/2026 12:25:42 PM	rw-r--r--	n.yengalareddy	
kraken_out		6/24/2026 12:26:13 PM	rw-r--r--	n.yengalareddy	
mash_output		6/24/2026 12:26:00 PM	rw-r--r--	n.yengalareddy	
mlst		6/24/2026 1:20:57 PM	rw-r--r--	n.yengalareddy	
multiqc		6/24/2026 1:09:47 PM	rw-r--r--	n.yengalareddy	
plasmid		6/24/2026 1:22:02 PM	rw-r--r--	n.yengalareddy	
serotypefinder		6/24/2026 1:21:22 PM	rw-r--r--	n.yengalareddy	
trimmomatic		6/24/2026 12:24:04 PM	rw-r--r--	n.yengalareddy	

J4-FL-M08815-240126.fasta ecoli_achtman_4 10 adk(10) fumC(11) gyrB(4) icd(8) mdh(8) purA(8) recA(2)

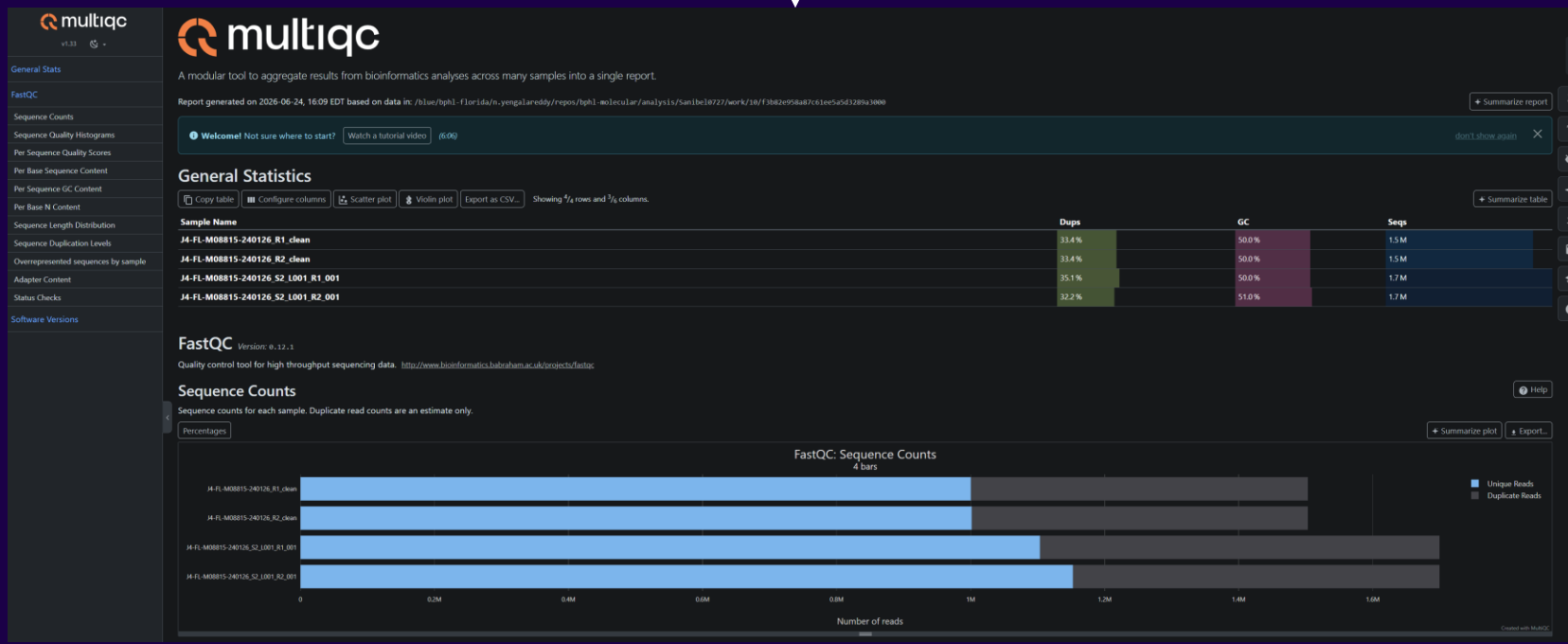
APPLICATION CONT.

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/Sanibel0727/output-20260624162501/J4-FL-M08815-240126/multiqc/

Name	Size	Changed	Rights	Owner
multiqc_report.html	2,29...	6/24/2026 1:09:47 PM	rw-r--r--	n.yengalareddy

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/Sanibel0727/output-20260624162501/J4-FL-M08815-240126/

Name	Size	Changed	Rights	Owner
..		6/24/2026 1:25:01 PM	rwxr-sr-x	n.yengalareddy
amrfinder		6/24/2026 1:21:03 PM	rwxr-sr-x	n.yengalareddy
assembly		6/24/2026 1:23:50 PM	rwxr-sr-x	n.yengalareddy
bbtools		6/24/2026 12:25:22 PM	rwxr-sr-x	n.yengalareddy
fastqc		6/24/2026 12:24:18 PM	rwxr-sr-x	n.yengalareddy
fastqc2		6/24/2026 12:25:42 PM	rwxr-sr-x	n.yengalareddy
kraken_out		6/24/2026 12:26:13 PM	rwxr-sr-x	n.yengalareddy
mask_output		6/24/2026 12:26:00 PM	rwxr-sr-x	n.yengalareddy
mlst		6/24/2026 1:20:57 PM	rwxr-sr-x	n.yengalareddy
multiqc		6/24/2026 1:09:47 PM	rwxr-sr-x	n.yengalareddy
plasmid		6/24/2026 1:22:02 PM	rwxr-sr-x	n.yengalareddy
serotypefinder		6/24/2026 1:21:22 PM	rwxr-sr-x	n.yengalareddy
trimmomatic		6/24/2026 12:24:04 PM	rwxr-sr-x	n.yengalareddy



APPLICATION CONT.

./../bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/Sanibel0727/output-20260624162501/J4-FL-M08815-240126/plasmid/

Name	Size	Changed	Rights	Owner
results_J4-FL-M088...	2 KB	6/24/2026 1:22:02 PM	rw-r--r--	n.yengalareddy

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/Sanibel0727/output-20260624162501/J4-FL-M08815-240126/

Name	Size	Changed	Rights	Owner
..		6/24/2026 1:25:01 PM	rw-r--r--	n.yengalareddy
amrfinder		6/24/2026 1:21:03 PM	rw-r--r--	n.yengalareddy
assembly		6/24/2026 1:23:50 PM	rw-r--r--	n.yengalareddy
bbtools		6/24/2026 12:25:22 PM	rw-r--r--	n.yengalareddy
fastqc		6/24/2026 12:24:18 PM	rw-r--r--	n.yengalareddy
fastqc2		6/24/2026 12:25:42 PM	rw-r--r--	n.yengalareddy
kraken_out		6/24/2026 12:26:13 PM	rw-r--r--	n.yengalareddy
mash_output		6/24/2026 12:26:00 PM	rw-r--r--	n.yengalareddy
mlst		6/24/2026 1:20:57 PM	rw-r--r--	n.yengalareddy
multic		6/24/2026 1:09:47 PM	rw-r--r--	n.yengalareddy
plasmid		6/24/2026 1:22:02 PM	rw-r--r--	n.yengalareddy
serotypefinder		6/24/2026 1:21:22 PM	rw-r--r--	n.yengalareddy
trimmomatic		6/24/2026 12:24:04 PM	rw-r--r--	n.yengalareddy

```
{
  "type": "software_result",
  "databases": {
    "PlasmidFinder-2.2.0": {
      "type": "database",
      "database_name": "PlasmidFinder",
      "database_version": "2.2.0",
      "key": "PlasmidFinder-2.2.0",
      "database_commit": "unknown"
    },
    "seq_regions": {
      "IncB/O/K/Z_3_GQ259888": {
        "type": "seq_region",
        "phenotypes": [],
        "ref_database": "PlasmidFinder-2.2.0:enterobacteriales",
        "key": "IncB/O/K/Z_3_GQ259888",
        "gene": true,
        "name": "IncB/O/K/Z",
        "identity": 99.34,
        "alignment_length": 152,
        "ref_seq_length": 152,
        "coverage": 100.0,
        "depth": 150.94,
        "ref_id": "IncB/O/K/Z_3_GQ259888",
        "ref_acc": "GQ259888",
        "ref_start_pos": 1,
        "ref_end_pos": 153,
        "query_id": "NA",
        "query_start_pos": "NA",
        "query_end_pos": "NA",
        "note": "",
        "query_string": "",
        "alignment_string": "",
        "ref_string": "",
        "seq_variations": {},
        "phenotypes": {}
      },
      "software_executions": {
        "2b47870f449918b379d2b12f038eef74e7abb99a": {
          "type": "software_exec",
          "command": "/usr/local/lib/python3.12/dist-packages/plasmidfinder/__main__.py -i J4-FL-M08815-240126_R1_clean.fastq.gz J4-FL-M08815-240126_R2_clean.fastq.gz -o ./",
          "parameters": {
            "infile": ["J4-FL-M08815-240126_R1_clean.fastq.gz", "J4-FL-M08815-240126_R2_clean.fastq.gz"],
            "outdir": "./",
            "tmp_dir": null,
            "method_path": null,
            "db_path": "/plasmidfinder_db",
            "databases": null,
            "min_cov": 0.6,
            "threshold": 0.9,
            "extended_output": false,
            "speciesinfo_json": null,
            "quiet": false,
            "out_json": null,
            "legacy": false
          },
          "key": "2b47870f449918b379d2b12f038eef74e7abb99a"
        }
      },
      "aln_hits": {}
    },
    "software_name": "PlasmidFinder",
    "software_version": "3.0.3",
    "software_commit": "unknown",
    "run_date": "24.06.2026",
    "key": "PlasmidFinder-3.0.3"
  }
}
```

APPLICATION CONT.

/.../n.yengalareddy/repos/bphl-molecular/analysis/Sanibel0727/output-20260624162501/J4-FL-M08815-240126/serotypefinder/

Name	Size	Changed	Rights	Owner
..		6/24/2026 1:22:02 PM	rw-r--r--	n.yengalareddy
data.json	2 KB	6/24/2026 1:21:22 PM	rw-r--r--	n.yengalareddy

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/Sanibel0727/output-20260624162501/J4-FL-M08815-240126/

Name	Size	Changed	Rights	Owner
..		6/24/2026 1:25:01 PM	rw-r--r--	n.yengalareddy
amrfinder		6/24/2026 1:21:03 PM	rw-r--r--	n.yengalareddy
assembly		6/24/2026 1:23:50 PM	rw-r--r--	n.yengalareddy
bbtools		6/24/2026 12:25:22 PM	rw-r--r--	n.yengalareddy
fastqc		6/24/2026 12:24:18 PM	rw-r--r--	n.yengalareddy
fastqc2		6/24/2026 12:25:42 PM	rw-r--r--	n.yengalareddy
kraken_out		6/24/2026 12:26:13 PM	rw-r--r--	n.yengalareddy
mash_output		6/24/2026 12:26:00 PM	rw-r--r--	n.yengalareddy
mlst		6/24/2026 1:20:57 PM	rw-r--r--	n.yengalareddy
multiqc		6/24/2026 1:09:47 PM	rw-r--r--	n.yengalareddy
plasmid		6/24/2026 1:22:02 PM	rw-r--r--	n.yengalareddy
serotypefinder		6/24/2026 1:21:22 PM	rw-r--r--	n.yengalareddy
trimmomatic		6/24/2026 12:24:04 PM	rw-r--r--	n.yengalareddy

```
{"serotypefinder": {"user_input": {"filename(s)": ["J4-FL-M08815-240126_R1_clean.fastq.gz", "J4-FL-M08815-240126_R2_clean.fastq.gz"], "method": "kma", "file_format": "fastq"}, "run_info": {"date": "24.06.2026", "time": "16:21:22"}, "results": {"H_type": {"fliC_52_AJ605764_H4": {"gene": "fliC", "serotype": "H4", "identity": 100.0, "HSP_length": 1050, "template_length": 1050, "position_in_ref": "1..1051", "contig_name": "NA", "positions_in_contig": "NA..NA", "accession": "AJ605764", "coverage": 100.0, "hit_id": "fliC_52_AJ605764_H4"}}, "O_type": {"wzx_37_EU294173_O29": {"gene": "wzx", "serotype": "O29", "identity": 99.93, "HSP_length": 1401, "template_length": 1401, "position_in_ref": "1..1402", "contig_name": "NA", "positions_in_contig": "NA..NA", "accession": "EU294173", "coverage": 100.0, "hit_id": "wzx_37_EU294173_O29"}, "wzy_23_EU294173_O29": {"gene": "wzy", "serotype": "O29", "identity": 100.0, "HSP_length": 1164, "template_length": 1164, "position_in_ref": "1..1165", "contig_name": "NA", "positions_in_contig": "NA..NA", "accession": "EU294173", "coverage": 100.0, "hit_id": "wzy_23_EU294173_O29"}}}}}
```

APPLICATION CONT.

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/Sanibel0727/output-20260624162501/J4-FL-M08815-240126/

Name	Size	Changed	Rights	Owner
..		6/24/2026 1:25:01 PM	rwxr-sr-x	n.yengalareddy
amrfinder		6/24/2026 1:21:03 PM	rwxr-sr-x	n.yengalareddy
assembly		6/24/2026 1:23:50 PM	rwxr-sr-x	n.yengalareddy
bbtools		6/24/2026 12:25:22 PM	rwxr-sr-x	n.yengalareddy
fastqc		6/24/2026 12:24:18 PM	rwxr-sr-x	n.yengalareddy
fastqc2		6/24/2026 12:25:42 PM	rwxr-sr-x	n.yengalareddy
kraken_out		6/24/2026 12:26:13 PM	rwxr-sr-x	n.yengalareddy
mash_output		6/24/2026 12:26:00 PM	rwxr-sr-x	n.yengalareddy
mlst		6/24/2026 1:20:57 PM	rwxr-sr-x	n.yengalareddy
multiqc		6/24/2026 1:09:47 PM	rwxr-sr-x	n.yengalareddy
plasmid		6/24/2026 1:22:02 PM	rwxr-sr-x	n.yengalareddy
serotypefinder		6/24/2026 1:21:22 PM	rwxr-sr-x	n.yengalareddy
trimmomatic		6/24/2026 12:24:04 PM	rwxr-sr-x	n.yengalareddy



/.../n.yengalareddy/repos/bphl-molecular/analysis/Sanibel0727/output-20260624162501/J4-FL-M08815-240126/trimmomatic/

Name	Size	Changed	Rights	Owner
..		6/24/2026 1:22:02 PM	rwxr-sr-x	n.yengalareddy
J4-FL-M08815-240...	210,...	6/24/2026 12:24:05 PM	rw-r--r--	n.yengalareddy
J4-FL-M08815-240...	220,...	6/24/2026 12:24:05 PM	rw-r--r--	n.yengalareddy

CONCLUSION



Fundamentals of
Sanibel



Installation and setup
of Sanibel in HPG



Successfully executed
job query for Sanibel



Generated output files

Sanibel v2.1.0 (currently in testing phase)

- New species ID workflow based on whole genome alignment (ANI) + new contamination flag
- New QC fields for the summary report + AMR results
- New interactive dashboard report (html)
- New serotyping module for *Listeria monocytogenes* (LisSero)
- Compatible with Nextflow v26 and backward compatible with previous versions



Advanced Molecular Detection

Southeast Region Bioinformatics

Questions?

bphl-sebioinformatics@flhealth.gov

Molly Mitchell, PhD

Bioinformatics Supervisor

Molly.Mitchell@flhealth.gov

Nikhil Reddy, MS

Bioinformatician

Nikhil.Yengala@flhealth.gov

Sam Bernhoft, MPH

Bioinformatician

Samantha.bernhoft@flhealth.gov

Arnold RodriguezHilario, MS

Bioinformatician

Arnold.RodriguezHilario@flhealth.gov